

复变函数与数理方程作业题答案

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1 复变函数与积分变换

1.1 第 01 次课作业

本次课程不安排作业。

1.2 第 02 次课作业

习题 1.1 下列式子在复平面上具有怎样的意义？(并给出具体过程)

$$(1) |z - a| = |z - b| \quad (a, b \text{ 为复常数}), \quad (2) |z| + \operatorname{Re} z \leq 1, \\ (3) \operatorname{Re} \left(\frac{1}{z} \right) = 2, \quad (4) \left| \frac{z-1}{z+1} \right| = 2, \quad (5) \arg(z - i) = \frac{\pi}{4}.$$

解：(1) 设

$$z = x + iy, \quad a = a_1 + ia_2, \quad b = b_1 + ib_2,$$

其中 $x, y, a_1, a_2, b_1, b_2 \in \mathbb{R}$ 。原方程等价于

$$(x - a_1)^2 + (y - a_2)^2 = (x - b_1)^2 + (y - b_2)^2,$$

整理得

$$(a_1 - b_1) \left(x - \frac{a_1 + b_1}{2} \right) + (a_2 - b_2) \left(y - \frac{a_2 + b_2}{2} \right) = 0.$$

因此原方程在复平面是一条经过 a 、 b 线段中点且与该线段垂直的直线。

(2) 设

$$z = x + iy, \quad x, y \in \mathbb{R}.$$

代入原等式得

$$\sqrt{x^2 + y^2} + x \leq 1,$$

整理得

$$y^2 \leq -2 \left(x - \frac{1}{2} \right).$$

因此原方程在复平面是开口向左、顶点在 $(\frac{1}{2}, 0)$ 的抛物线及其包围区域内部。

(3) 设

$$z = x + iy, \quad x, y \in \mathbb{R}.$$

由于

$$\frac{1}{z} = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2},$$

原方程为

$$\frac{x}{x^2 + y^2} = 2 \iff \left(x - \frac{1}{4} \right)^2 + y^2 = \frac{1}{16}, \quad x^2 + y^2 \neq 0.$$

因此原方程在复平面是以 $(\frac{1}{4}, 0)$ 为圆心、 $\frac{1}{4}$ 为半径且挖去原点的圆。

(4) 设

$$z = x + iy, \quad x, y \in \mathbb{R}.$$

代入原式得

$$\sqrt{(x-1)^2 + y^2} = 2\sqrt{(x+1)^2 + y^2}, \quad (x+1)^2 + y^2 \neq 0,$$

整理得到

$$\left(x + \frac{5}{3} \right)^2 + y^2 = \frac{16}{9}.$$

因此原方程在复平面是以 $(-\frac{5}{3}, 0)$ 为圆心、 $\frac{4}{3}$ 为半径的圆。

(5) 设

$$z = x + iy, \quad x, y \in \mathbb{R}.$$

原式即

$$\arg(x + i(y-1)) = \frac{\pi}{4} \iff \frac{y-1}{x} = \tan \frac{\pi}{4} = 1, \quad x > 0.$$

故原方程在复平面是以 $(0, 1)$ 为端点向右上方的一条射线 (不包括端点)。□

习题 1.2 计算下列数值 (a, b, φ 为实常数)

$$(1) (1+i)^i, \quad (2) \cos \varphi + \cos 2\varphi + \cdots + \cos n\varphi, \quad (3) \sin(a+ib).$$

解: (1) 我们有

$$(1+i)^i = e^{i \ln(1+i)} = e^{i(\ln \sqrt{2} + i(\frac{\pi}{4} + 2k\pi))} = e^{-\frac{\pi}{4} - 2k\pi} \left(\cos(\ln \sqrt{2}) + i \sin(\ln \sqrt{2}) \right).$$

其中 $k \in \mathbb{Z}$ 。

(2) 我们有

$$\begin{aligned}\sum_{k=1}^n \cos k\varphi &= \sum_{k=1}^n \frac{e^{ik\varphi} + e^{-ik\varphi}}{2} = \frac{e^{i\varphi} - e^{i(n+1)\varphi}}{2(1 - e^{i\varphi})} + \frac{e^{-i\varphi} - e^{-i(n+1)\varphi}}{2(1 - e^{-i\varphi})} \\ &= \frac{1}{2} \left(\frac{e^{i(n+\frac{1}{2})\varphi} - e^{\frac{1}{2}i\varphi}}{e^{\frac{1}{2}i\varphi} - e^{-\frac{1}{2}i\varphi}} + \frac{e^{-\frac{1}{2}i\varphi} - e^{-i(n+\frac{1}{2})\varphi}}{e^{\frac{1}{2}i\varphi} - e^{-\frac{1}{2}i\varphi}} \right) = -\frac{1}{2} + \frac{1}{2} \frac{e^{i(n+\frac{1}{2})\varphi} - e^{-i(n+\frac{1}{2})\varphi}}{e^{i\frac{\varphi}{2}} - e^{-i\frac{\varphi}{2}}} \\ &= -\frac{1}{2} + \frac{\sin(n + \frac{1}{2})\varphi}{2 \sin \frac{\varphi}{2}}.\end{aligned}$$

(3) 我们有

$$\sin(a + ib) = \frac{e^{i(a+ib)} - e^{-i(a+ib)}}{2i} = \frac{i}{2} (e^b e^{-ia} - e^{-b} e^{ia}) = \frac{1}{2} (e^b + e^{-b}) \sin a + \frac{i}{2} (e^b - e^{-b}) \cos a. \quad \square$$

习题 1.3 试证明 $\overline{e^z} = e^{\bar{z}}$.

证明： 设

$$z = x + iy, \quad x, y \in \mathbb{R}.$$

则有

$$\overline{e^z} = \overline{e^{x+iy}} = \overline{e^x (\cos y + i \sin y)} = e^x (\cos y - i \sin y) = e^{x-iy} = e^{\bar{z}}. \quad \square$$

习题 1.4

(1) 用复变量表示过点 $1 + 3i$ 和 $-1 + 4i$ 直线的方程;

(2) 设 $A, C \in \mathbb{R}$, $B \in \mathbb{C}$, 问方程 $Azz^* + Bz^* + B^*z + C = 0$ 在什么条件是圆方程, 并求其圆心和半径.

解： (1) 经过 $(1, 3)$ 和 $(-1, 4)$ 的直线方程为

$$x + 2y - 7 = 0.$$

令

$$x = \frac{1}{2}(z + z^*), \quad y = \frac{1}{2i}(z - z^*),$$

代入上式, 则有

$$\left(\frac{1}{2} - i\right)z + \left(\frac{1}{2} + i\right)z^* - 7 = 0.$$

(2) 设

$$z = x + iy, \quad B = b_1 + ib_2.$$

其中 $x, y, b_1, b_2 \in \mathbb{R}$. 首先 $A \neq 0$, 否则原方程为直线方程. 将上式代入原方程, 可得

$$\left(x + \frac{b_1}{A}\right)^2 + \left(y + \frac{b_2}{A}\right)^2 = \frac{b_1^2 + b_2^2 - AC}{A^2}.$$

因此原方程为圆方程的充要条件为

$$A \neq 0, \quad BB^* - AC > 0,$$

且圆心为 $-\frac{B}{A}$, 半径为 $\frac{\sqrt{BB^* - AC}}{|A|}$. $\square$

习题 1.5 设 $f(z) = u(r, \varphi) + i v(r, \varphi)$ ($z = r e^{i\varphi}$), 试推导出极坐标下的 $C-R$ 条件, 并证明

$$f'(z) = \frac{r}{z} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right).$$

解: 我们有极坐标和直角坐标间的变换关系

$$\begin{aligned} x &= r \cos \varphi, \quad y = r \sin \varphi, \\ r &= \sqrt{x^2 + y^2}, \quad \varphi = \arctan \frac{y}{x}. \end{aligned}$$

因此有

$$\begin{aligned} \frac{\partial r}{\partial x} &= \frac{x}{\sqrt{x^2 + y^2}} = \frac{x}{r} = \cos \varphi, & \frac{\partial r}{\partial y} &= \frac{y}{\sqrt{x^2 + y^2}} = \frac{y}{r} = \sin \varphi \\ \frac{\partial \varphi}{\partial x} &= \frac{-\frac{y}{x^2}}{1 + \left(\frac{y}{x}\right)^2} = \frac{-y}{r^2} = -\frac{\sin \varphi}{r}, & \frac{\partial \varphi}{\partial y} &= \frac{\frac{1}{x}}{1 + \left(\frac{y}{x}\right)^2} = \frac{x}{r^2} = \frac{\cos \varphi}{r}. \end{aligned}$$

根据链式法则

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial x} = \cos \varphi \frac{\partial u}{\partial r} - \frac{\sin \varphi}{r} \frac{\partial u}{\partial \varphi}, & \frac{\partial u}{\partial y} &= \frac{\partial u}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial y} = \sin \varphi \frac{\partial u}{\partial r} + \frac{\cos \varphi}{r} \frac{\partial u}{\partial \varphi}, \\ \frac{\partial v}{\partial x} &= \frac{\partial v}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial v}{\partial \varphi} \frac{\partial \varphi}{\partial x} = \cos \varphi \frac{\partial v}{\partial r} - \frac{\sin \varphi}{r} \frac{\partial v}{\partial \varphi}, & \frac{\partial v}{\partial y} &= \frac{\partial v}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial v}{\partial \varphi} \frac{\partial \varphi}{\partial y} = \sin \varphi \frac{\partial v}{\partial r} + \frac{\cos \varphi}{r} \frac{\partial v}{\partial \varphi}. \end{aligned}$$

将它们代入直角坐标系中的 $C-R$ 条件

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y},$$

可得

$$\cos \varphi \frac{\partial u}{\partial r} - \frac{\sin \varphi}{r} \frac{\partial u}{\partial \varphi} = \sin \varphi \frac{\partial v}{\partial r} + \frac{\cos \varphi}{r} \frac{\partial v}{\partial \varphi}, \quad (1.1)$$

$$\sin \varphi \frac{\partial u}{\partial r} + \frac{\cos \varphi}{r} \frac{\partial u}{\partial \varphi} = -\cos \varphi \frac{\partial v}{\partial r} + \frac{\sin \varphi}{r} \frac{\partial v}{\partial \varphi}. \quad (1.2)$$

考虑 $(1.1) \times \cos \varphi + (1.2) \times \sin \varphi$ 和 $(1.1) \times \sin \varphi - (1.2) \times \cos \varphi$ 得到极坐标系下的 $C-R$ 条件

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \varphi}, \quad \frac{\partial u}{\partial \varphi} = -r \frac{\partial v}{\partial r}.$$

进一步的, 我们有

$$\begin{aligned} f'(z) &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \left(\cos \varphi \frac{\partial u}{\partial r} - \frac{\sin \varphi}{r} \frac{\partial u}{\partial \varphi} \right) + i \left(\cos \varphi \frac{\partial v}{\partial r} - \frac{\sin \varphi}{r} \frac{\partial v}{\partial \varphi} \right) \\ &= \left(\cos \varphi \frac{\partial u}{\partial r} + \sin \varphi \frac{\partial v}{\partial r} \right) + i \left(\cos \varphi \frac{\partial v}{\partial r} - \sin \varphi \frac{\partial u}{\partial r} \right) = \frac{r}{z} \left(\frac{\partial u}{\partial r} + i \frac{\partial v}{\partial r} \right). \quad \square \end{aligned}$$

习题 1.6 判断下列函数的可微性及解析性

$$(1) f(z) = xy^2 + ix^2y, \quad (2) f(z) = 2x^3 + i3y^3.$$

解: (1) 设 $f(z) = u + iv$, 其中

$$u(x, y) = xy^2, \quad v(x, y) = x^2y.$$

可知 u, v 在整个复平面上连续可微, 进一步的, 根据 $C-R$ 条件

$$\frac{\partial u}{\partial x} = y^2 \neq x^2 = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = 2xy \neq -2xy = -\frac{\partial u}{\partial y},$$

当且仅当 $x = y = 0$ 等号成立, 因此 $f(z)$ 只在原点处可微, 并且在复平面任一点不解析。

(2) 令

$$u(x, y) = 2x^3, \quad v(x, y) = 3y^3.$$

可知 u, v 在整个复平面上连续可微, 进一步的, 根据 $C-R$ 条件

$$\frac{\partial u}{\partial x} = 6x^2 \neq 9y^2 = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = 0 = -\frac{\partial u}{\partial y},$$

当且仅当 $y = \pm \frac{\sqrt{6}}{3}x$ 等号成立, 因此 $f(z)$ 只在直线 $y = \frac{\sqrt{6}}{3}x$ 和 $y = -\frac{\sqrt{6}}{3}x$ 上可微, 并且在复平面任一点不解析。□

习题 1.7 已知解析函数 $f(z)$ 的实部 $u(x, y)$ 或虚部 $v(x, y)$, 求该解析函数

$$(1) u = e^x \sin y, \quad (2) u = \frac{x^2 - y^2}{(x^2 + y^2)^2}, \quad f(\infty) = 0, \quad (3) u = \ln \rho, \quad f(1) = 0.$$

解: (1) 首先有

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = e^x \sin y - e^x \sin y = 0,$$

利用 $C-R$ 条件, 虚部 $v(x, y)$ 的偏导数满足

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = -e^x \cos y, \quad \frac{\partial v}{\partial y} = -\frac{\partial u}{\partial x} = e^x \sin y.$$

由此给出 $v(x, y)$ 的全微分

$$dv = -e^x \cos y dx + e^x \sin y dy = d(-e^x \cos y),$$

因此

$$v(x, y) = -e^x \cos y + C.$$

这里 $C \in \mathbb{R}$ 为任意常数, 从而有

$$f(z) = e^x \sin y - ie^x \cos y + iC = -ie^z + iC.$$

(2) 将实部 $u(x, y)$ 写成极坐标下的形式

$$u(\rho, \varphi) = \frac{\cos 2\varphi}{\rho^2},$$

在极坐标下 $u(\rho, \varphi)$ 满足

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} = 4\rho^{-4} \cos 2\varphi - 4\rho^{-4} \cos 2\varphi = 0.$$

是调和函数。利用 $C-R$ 条件，虚部 $v(\rho, \varphi)$ 的偏导数满足

$$\frac{\partial v}{\partial \varphi} = \rho \frac{\partial u}{\partial \rho} = -2\rho^{-2} \cos 2\varphi, \quad \frac{\partial v}{\partial \rho} = -\rho^{-1} \frac{\partial u}{\partial \varphi} = 2\rho^{-3} \sin 2\varphi.$$

由此给出 $v(\rho, \varphi)$ 的全微分

$$dv = 2\rho^{-3} \sin 2\varphi d\rho - 2\rho^{-2} \cos 2\varphi d\varphi = d\left(-\frac{\sin 2\varphi}{\rho^2}\right),$$

因此

$$v = -\frac{\sin 2\varphi}{\rho^2} + C,$$

这里 $C \in \mathbb{R}$ 为任意常数，从而有

$$f(z) = \frac{\cos 2\varphi}{\rho^2} - i \frac{\sin 2\varphi}{\rho^2} + iC = \frac{1}{z^2} + iC.$$

再根据条件 $f(\infty) = 0$ ，可得 $C = 0$ ，因此

$$f(z) = \frac{1}{z^2}.$$

(3) 在极坐标下 $u(\rho, \varphi)$ 满足

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \varphi^2} = 0.$$

是调和函数。利用 $C-R$ 条件，虚部 $v(\rho, \varphi)$ 的偏导数满足

$$\frac{\partial v}{\partial \varphi} = \rho \frac{\partial u}{\partial \rho} = 1, \quad \frac{\partial v}{\partial \rho} = -\rho^{-1} \frac{\partial u}{\partial \varphi} = 0.$$

由此给出 $v(\rho, \varphi)$ 的全微分

$$dv = d\varphi,$$

因此

$$v = \varphi + C,$$

这里 $C \in \mathbb{R}$ 为任意常数，从而有

$$f(z) = \ln \rho + i\varphi + iC = \ln z + iC.$$

再根据条件 $f(1) = 0$ ，可得 $C = 0$ ，因此

$$f(z) = \ln z. \quad \square$$

1.3 第 03 次课作业

习题 1.8 计算 $\operatorname{Re} z$ 沿曲线 $\ell: z = t + it^2$ ($0 \leq t \leq 1$) 的积分。

解: 首先有 $dz = (1 + 2it)dt$, $\operatorname{Re} z = t$, 从而积分为

$$\int_{\ell} \operatorname{Re} z \, ds = \int_0^1 t(1 + 2it) \, dt = \frac{1}{2} + \frac{2}{3}i. \quad \square$$

习题 1.9 (1) 已知函数 $\psi(t, x) = e^{2tx - t^2}$, 将 x 作为参数, t 为复变数, 试应用柯西积分公式将 $\frac{\partial^n \psi}{\partial t^n} \Big|_{t=0}$ 表示为回路积分。(2) 对回路积分进行变量替换 $\zeta = x - z$, 并由此证明

$$\frac{\partial^n \psi}{\partial t^n} \Big|_{t=0} = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}.$$

解: (1) 把 $\frac{\partial^n \psi}{\partial t^n} \Big|_{t=0}$ 表为回路积分如下

$$\frac{\partial^n \psi}{\partial t^n} \Big|_{t=0} = \frac{n!}{2\pi i} \oint_{\ell} \frac{e^{2\zeta x - \zeta^2}}{(\zeta - t)^{n+1}} \, d\zeta \Big|_{t=0} = \frac{n!}{2\pi i} \oint_{\ell} \frac{e^{2\zeta x - \zeta^2}}{\zeta^{n+1}} \, d\zeta.$$

(2) 将 $\zeta = x - z$ 代入上式得

$$\frac{\partial^n \psi}{\partial t^n} \Big|_{t=0} = \frac{n!}{2\pi i} \oint_{\ell} \frac{e^{x^2 - z^2}}{(x - z)^{n+1}} \, d(-z) = (-1)^n e^{x^2} \frac{n!}{2\pi i} \oint_{\ell} \frac{e^{-z^2}}{(z - x)^{n+1}} \, dz = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}. \quad \square$$

习题 1.10 计算以下积分

$$\begin{aligned} (1) & \oint_{|z|=1} \frac{dz}{\cos z}, \quad (2) \oint_{|z|=1} \frac{dz}{z^2 + 2z + 4}, \quad (3) \oint_{|z|=2} z^3 \cos z \, dz \\ (4) & \oint_{|z|=2} \frac{\sin z}{(z - \pi/2)^2} \, dz, \quad (5) \oint_{|z|=4} \left(\frac{4}{z+1} + \frac{3}{z+2i} \right) \, dz. \end{aligned}$$

解: (1) 由于被积函数 $f(z) = \frac{1}{\cos z}$ 在单位圆内解析, 利用柯西积分定理可知

$$\oint_{|z|=1} \frac{dz}{\cos z} = 0.$$

(2) 由于被积函数 $f(z) = \frac{1}{z^2 + 2z + 4}$ 在单位圆内解析, 利用柯西积分定理可知

$$\oint_{|z|=1} \frac{dz}{z^2 + 2z + 4} = 0.$$

(3) 由于被积函数 $f(z) = z^3 \cos z$ 在 $|z| \leq 2$ 内解析, 利用柯西积分定理可知

$$\oint_{|z|=2} z^3 \cos z \, dz = 0.$$

(4) 由于被积函数 $f(z) = \frac{\sin z}{(z - \pi/2)^2}$ 在 $|z| \leq 2$ 内有奇点 $\pi/2$, 利用柯西积分公式可知

$$\oint_{|z|=2} \frac{\sin z}{(z - \pi/2)^2} dz = \frac{2\pi i}{1!} \frac{d \sin z}{dz} \bigg|_{z=\pi/2} = 2\pi i \cos(\pi/2) = 0.$$

(5) 由于被积函数 $f(z) = \frac{4}{z+1} + \frac{3}{z+2i}$ 在 $|z| \leq 4$ 内有奇点 $-1, -2i$, 利用柯西积分公式可知

$$\oint_{|z|=4} \left(\frac{4}{z+1} + \frac{3}{z+2i} \right) dz = \oint_{|z|=4} \frac{4}{z+1} dz + \oint_{|z|=4} \frac{3}{z+2i} dz = 14\pi i. \square$$

习题 1.11 用积分

$$\oint_{|z|=1} \frac{dz}{z+2},$$

计算积分

$$\int_0^\pi \frac{1+2\cos\theta}{5+4\cos\theta} d\theta.$$

解: 由于被积函数 $f(z) = \frac{1}{z+2}$ 在单位圆内解析, 利用柯西积分定理可知

$$\oint_{|z|=1} \frac{dz}{z+2} = 0.$$

令 $z = \cos\theta + i\sin\theta$, $0 \leq \theta \leq 2\pi$, 可得

$$\begin{aligned} \oint_{|z|=1} \frac{dz}{z+2} &= \int_0^{2\pi} \frac{-\sin\theta + i\cos\theta}{\cos\theta + i\sin\theta + 2} d\theta = \int_0^{2\pi} \frac{(-\sin\theta + i\cos\theta)(2 + \cos\theta - i\sin\theta)}{5 + 4\cos\theta} d\theta \\ &= \int_0^{2\pi} \frac{-2\sin\theta + i(1+2\cos\theta)}{5+4\cos\theta} d\theta = \int_0^{2\pi} \frac{-2\sin\theta}{5+4\cos\theta} d\theta + i \int_0^{2\pi} \frac{1+2\cos\theta}{5+4\cos\theta} d\theta. \end{aligned}$$

所以积分

$$\int_0^{2\pi} \frac{1+2\cos\theta}{5+4\cos\theta} d\theta = 0.$$

注意到被积函数是 2π 周期的, 且为偶函数, 所以

$$\int_0^\pi \frac{1+2\cos\theta}{5+4\cos\theta} d\theta = \frac{1}{2} \int_0^{2\pi} \frac{1+2\cos\theta}{5+4\cos\theta} d\theta = 0. \square$$

习题 1.12 求下列幂级数的收敛圆

$$\begin{aligned} (1) \sum_{k=1}^{\infty} \frac{1}{k} (z-i)^k, \quad (2) \sum_{k=1}^{\infty} k^{\ln k} (z-2)^k, \\ (3) \sum_{k=1}^{\infty} k! \left(\frac{z}{k}\right)^k, \quad (4) \sum_{k=1}^{\infty} k^k (z-3)^k. \end{aligned}$$

解: (1) 收敛半径为

$$R = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{1/k}{1/(k+1)} \right| = \lim_{k \rightarrow \infty} \left| \frac{k+1}{k} \right| = 1.$$

以上极限存在, 所以收敛圆为 $|z - i| < 1$.

(2) 收敛半径为

$$R = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{|a_k|}} = \lim_{k \rightarrow \infty} k^{-\frac{\ln k}{k}} = \lim_{k \rightarrow \infty} e^{-\frac{(\ln k)^2}{k}} = 1.$$

以上极限存在, 所以收敛圆为 $|z - 2| < 1$.

(3) 收敛半径为

$$R = \lim_{k \rightarrow \infty} \left| \frac{a_k}{a_{k+1}} \right| = \lim_{k \rightarrow \infty} \left| \frac{k!(k+1)^{k+1}}{k^k(k+1)!} \right| = \lim_{k \rightarrow \infty} \left(1 + \frac{1}{k} \right)^k = e.$$

以上极限存在, 所以收敛圆为 $|z| < e$.

(4) 收敛半径为

$$R = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{|a_k|}} = \lim_{k \rightarrow \infty} \frac{1}{\sqrt[k]{|k^k|}} = \lim_{k \rightarrow \infty} \frac{1}{k} = 0.$$

以上极限存在, 所以收敛圆为 $|z - 3| = 0$, 只要 $z \neq 3$ 幂级数就发散。□

习题 1.13 在指定点 z_0 的邻域上将下列函数展开为泰勒级数 (不必给出收敛半径)

$$\begin{aligned} (1) \sqrt[3]{z}, z_0 = i, \quad (2) \ln(1 + e^z), z_0 = 0, \quad (3) \int_0^z \frac{\sin \xi}{\xi} d\xi, z_0 = 0, \\ (4) \frac{1}{z^2}, z_0 = -1, \quad (5) \sin^2 z, z_0 = 0, \quad (6) \frac{2}{z^2 - 4z + 3}, z_0 = 2. \end{aligned}$$

解: (1) $f(z) = \sqrt[3]{z}$ 在 $z_0 = i$ 处展开为泰勒级数

$$\sqrt[3]{z} = \sqrt[3]{(z - i) + i} = i^{\frac{1}{3}}(1 - i(z - i))^{\frac{1}{3}} = i^{\frac{1}{3}} \left(1 + \sum_{k=1}^{\infty} \frac{\prod_{l=0}^{k-1} (\frac{1}{3} - l)}{k!} (-i)^k (z - i)^k \right).$$

注: $\sqrt[3]{z}$ 是多值函数, 这里只考虑其单值分支。

(2) 解法 1: 注意到

$$\begin{aligned} \ln(1 + e^z) &= e^z - \frac{e^{2z}}{2} + \frac{e^{3z}}{3} - \cdots + \frac{(-1)^{k+1} e^{kz}}{k} + \cdots \\ &= \left(1 - \frac{1}{2} + \frac{1}{3} + \cdots \right) + \left(z - \frac{1}{2}(2z) + \frac{1}{3}(3z) + \cdots \right) + \frac{1}{2!} \left(z^2 - \frac{1}{2}(2z)^2 + \cdots \right) + \cdots \\ &= \ln 2 + \sum_{k=1}^{\infty} \frac{(\sum_{i=1}^{\infty} (-1)^{i-1} \cdot i^{k-1})}{k!} z^k \end{aligned}$$

解法 2: $f(x) = \ln(1 + x)$ 在 $x = 1$ 处的泰勒展开为

$$\ln(1 + x) = \ln 2 + \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k \cdot 2^k} x^k$$

将 $x = e^z$ 带入, 并注意到 $e^{kz} = \sum_{n=0}^{\infty} \frac{(kz)^n}{n!}$ 可得:

$$\ln(1 + e^z) = \ln 2 + \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k \cdot 2^k} \left(\sum_{n=0}^{\infty} \frac{k^n}{n!} z^n \right) = \ln 2 + \sum_{n=0}^{\infty} \left(\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k \cdot 2^k} \frac{k^n}{n!} \right) z^n$$

(3) 注意到

$$\frac{\sin z}{z} = \frac{1}{z} \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!} = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k+1)!}.$$

由于 $\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$, 则 $z = 0$ 为可去奇点, 可以补充定义 $\frac{\sin z}{z}$ 在 $z = 0$ 出的值, 于是上式收敛范围为 $|z| < \infty$, 符合逐项积分条件, 所以 $\int_0^z \frac{\sin \xi}{\xi} d\xi$ 在 $z_0 = i$ 处展开为泰勒级数

$$\int_0^z \frac{\sin \xi}{\xi} d\xi = \int_0^z \sum_{k=0}^{\infty} (-1)^k \frac{\xi^{2k}}{(2k+1)!} d\xi = \sum_{k=0}^{\infty} (-1)^k \int_0^z \frac{\xi^{2k}}{(2k+1)!} d\xi = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)(2k+1)!}$$

(4) $f(z) = \frac{1}{z^2}$ 在 $z_0 = -1$ 处展开为泰勒级数

$$\frac{1}{z^2} = ((z+1) - 1)^{-2} = 1 + \sum_{k=1}^{\infty} \frac{(k+1)!}{k!} (z+1)^k = \sum_{k=0}^{\infty} (k+1)(z+1)^k$$

(5) $f(z) = \sin^2 z$ 在 $z_0 = 0$ 处展开为泰勒级数

$$\sin^2 z = \frac{1 - \cos 2z}{2} = \frac{1}{2} - \frac{1}{2} \sum_{k=0}^{\infty} (-1)^k \frac{(2z)^{2k}}{(2k)!} = \frac{1}{2} \sum_{k=1}^{\infty} (-1)^{k-1} \frac{(2z)^{2k}}{(2k)!}$$

(6) $f(z) = \frac{2}{z^2 - 4z + 3}$ 在 $z_0 = 2$ 处展开为泰勒级数

$$\begin{aligned} \frac{2}{z^2 - 4z + 3} &= \frac{1}{z-3} - \frac{1}{z-1} = \frac{1}{(z-2)-1} - \frac{1}{(z-2)+1} \\ &= \left(-\sum_{k=0}^{\infty} (z-2)^k - \sum_{k=0}^{\infty} (-1)^k (z-2)^k \right) = \sum_{k=0}^{\infty} [-1 - (-1)^k] (z-2)^k. \quad \square \end{aligned}$$

1.4 第 04 次课作业

习题 1.14 在挖去奇点 z_0 的环域上或指定环域上将下列函数展开为洛朗级数

- (1) $z^5 e^{1/z}$, $z_0 = 0$, (2) $\frac{1}{z^2(z-1)}$, $z_0 = 1$, (3) $\sin \frac{1}{1-z}$, $0 < |z-1| < \infty$,
 (4) $\frac{1}{z^2 - 3z + 2}$ 在 $1 < |z| < 2$ 或 $2 < |z| < \infty$, (5) $\sin \frac{1}{z}$ 在奇点,
 (6) $\frac{e^z}{z}$ 在奇点, (7) $\frac{1}{z^2(z^2-1)^2}$ 在 $0 < |z| < 1$ 或 $1 < |z| < \infty$.

解: (1) $f(z) = e^{\frac{1}{z}}$ 在挖去 $z_0 = 0$ 的环域上可展开洛朗级数:

$$e^{\frac{1}{z}} = \sum_{k=0}^{\infty} \frac{1}{k!} \left(\frac{1}{z}\right)^k = \sum_{k=-\infty}^0 \frac{1}{(-k)!} z^k,$$

因此 $z^5 e^{\frac{1}{z}}$ 在挖去 $z_0 = 0$ 的环域上可展开洛朗级数:

$$z^5 e^{\frac{1}{z}} = z^5 \sum_{k=-\infty}^0 \frac{1}{(-k)!} z^k = \sum_{k=-\infty}^0 \frac{1}{(-k)!} z^{k+5} = \sum_{k=-\infty}^5 \frac{1}{(-k+5)!} z^k.$$

(2) 由于

$$\frac{1}{z^2(z-1)} = \frac{1}{z-1} \left(-\frac{1}{z}\right)' = -\frac{1}{z-1} \left(\frac{1}{1-(1-z)}\right)',$$

而 $\frac{1}{1-(1-z)}$ 在挖去 $z_0 = 1$ 的环域上可展开洛朗级数:

$$\frac{1}{1-(1-z)} = \sum_{k=0}^{\infty} (1-z)^k.$$

因此 $f(z) = \frac{1}{z^2(z-1)}$ 在挖去 $z_0 = 1$ 的环域上可展开洛朗级数:

$$\begin{aligned} \frac{1}{z^2(z-1)} &= -\frac{1}{z-1} \left(\sum_{k=0}^{\infty} (1-z)^k\right)' = -\frac{1}{1-z} \sum_{k=1}^{\infty} k(1-z)^{k-1} \\ &= -\sum_{k=1}^{\infty} k(1-z)^{k-2} = -\sum_{k=-1}^{\infty} (k+2)(1-z)^k = \sum_{k=-1}^{\infty} (-1)^{(k+1)}(k+2)(z-1)^k. \end{aligned}$$

(3) 已知

$$\sin z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!},$$

而 $\sin \frac{1}{1-z}$ 的奇点为 $z = 1$, 因此在 $0 < |z-1| < \infty$ 内可展开洛朗级数:

$$f(z) = \sin \frac{1}{1-z} = \sum_{k=0}^{\infty} (-1)^k \frac{\left(\frac{1}{1-z}\right)^{2k+1}}{(2k+1)!} = \sum_{k=0}^{\infty} \frac{(-1)^{k+1}}{(2k+1)!(z-1)^{2k+1}} = \sum_{k=-\infty}^0 (-1)^{-k+1} \frac{(z-1)^{2k-1}}{(-2k+1)!}.$$

(4) $f(z) = \frac{1}{z^2-3z+2}$ 可化简为:

$$\frac{1}{z^2-3z+2} = \frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1},$$

下面根据题目要求在不同区域上做洛朗展开: (a) 在 $1 < |z| < 2$ 区域内

$$\begin{aligned} \frac{1}{z^2-3z+2} &= \frac{1}{z-2} - \frac{1}{z-1} = -\frac{1}{2(1-\frac{z}{2})} - \frac{1}{z(1-\frac{1}{z})} \\ &= -\frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{z}{2}\right)^k - \frac{1}{z} \sum_{k=0}^{\infty} \left(\frac{1}{z}\right)^k = -\sum_{k=0}^{\infty} \frac{z^k}{2^{k+1}} - \sum_{k=0}^{\infty} \frac{1}{z^{k+1}} = -\sum_{k=-\infty}^{-1} z^k - \sum_{k=0}^{\infty} \frac{z^k}{2^{k+1}}. \end{aligned}$$

(b) 在 $2 < |z| < \infty$ 区域内

$$\begin{aligned}\frac{1}{z^2 - 3z + 2} &= \frac{1}{z-2} - \frac{1}{z-1} = \frac{1}{z(1-\frac{2}{z})} - \frac{1}{z(1-\frac{1}{z})} \\ &= \frac{1}{z} \sum_{k=0}^{\infty} \left(\frac{2}{z}\right)^k - \sum_{k=0}^{\infty} \left(\frac{1}{z}\right)^{k+1} = \sum_{k=0}^{\infty} \frac{2^k}{z^{k+1}} - \sum_{k=0}^{\infty} \left(\frac{1}{z}\right)^{k+1} = \sum_{k=1}^{\infty} \frac{2^k - 1}{z^{k+1}} = \sum_{k=-\infty}^{-2} (2^{-k-1} - 1)z^k.\end{aligned}$$

(5) 由于

$$\sin z = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!},$$

而 $f(z) = \sin \frac{1}{z}$ 的奇点为 $z=0$, 因此 $\sin \frac{1}{z}$ 在挖去 $z_0=0$ 的环域上可展开洛朗级数:

$$\sin \frac{1}{z} = \sum_{k=0}^{\infty} (-1)^k \frac{\left(\frac{1}{z}\right)^{2k+1}}{(2k+1)!} = \sum_{k=-\infty}^0 (-1)^k \frac{z^{2k-1}}{(-2k+1)!}.$$

(6) 由于

$$e^z = \sum_{k=0}^{\infty} \frac{z^k}{k!},$$

而 $f(z) = \frac{e^z}{z}$ 的奇点为 $z_0=0$, 因此 $\frac{e^z}{z}$ 在挖去 $z_0=0$ 的环域上可展开洛朗级数:

$$\frac{e^z}{z} = \frac{\sum_{k=0}^{\infty} \frac{z^k}{k!}}{z} = \sum_{k=0}^{\infty} \frac{z^{k-1}}{k!} = \sum_{k=-1}^{\infty} \frac{z^k}{(k+1)!}.$$

(7) 下面根据题目要求在不同区域上做洛朗展开: (a) 在 $0 < |z| < 1$ 区域内

$$\frac{1}{z^2(z^2-1)^2} = \frac{1}{z^2(1-z^2)^2} = \frac{1}{z^2} \frac{1}{(1-z^2)^2},$$

利用幂级数在收敛圆内部可以逐项微分的性质, 可得

$$\frac{1}{(1-z)^2} = \left(\frac{1}{1-z}\right)' = \left(\sum_{k=0}^{\infty} z^k\right)' = \sum_{k=1}^{\infty} k z^{k-1}.$$

由此得到

$$\frac{1}{z^2(z^2-1)^2} = \frac{1}{z^2} \frac{1}{(1-z^2)^2} = \frac{1}{z^2} \sum_{k=1}^{\infty} k z^{2k-2} = \sum_{k=1}^{\infty} k z^{2k-4} = \sum_{k=-1}^{\infty} (k+2) z^{2k}.$$

(b) 在 $1 < |z| < \infty$ 内有:

$$\frac{1}{z^2(z^2-1)^2} = \frac{1}{z^6(1-\frac{1}{z^2})^2} = \frac{1}{z^6} \frac{1}{(1-\frac{1}{z^2})^2} = \frac{1}{z^6} \sum_{k=1}^{\infty} k \left(\frac{1}{z^2}\right)^{k-1} = \sum_{k=1}^{\infty} \frac{k}{z^{2k+4}} = \sum_{k=-\infty}^{-3} (-k-2) z^{2k}. \quad \square$$

习题 1.15 利用级数

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}, \quad |z| < 1,$$

证明实三角级数

$$\sum_{n=0}^{\infty} r^n \cos n\theta = \frac{1 - r \cos \theta}{1 - 2r \cos \theta + r^2}, \quad 0 < r < 1,$$

$$\sum_{n=0}^{\infty} r^n \sin n\theta = \frac{r \sin \theta}{1 - 2r \cos \theta + r^2}, \quad 0 < r < 1.$$

证明：令 $z = re^{i\theta}$, $r = |z| \in (0, 1)$, 则由级数

$$\sum_{n=0}^{\infty} z^n = \frac{1}{1 - z}, \quad |z| < 1,$$

可知

$$\sum_{n=0}^{\infty} r^n e^{in\theta} = \sum_{n=0}^{\infty} z^n = \frac{1}{1 - z} = \frac{1}{1 - re^{i\theta}},$$

即得,

$$\begin{aligned} \sum_{n=0}^{\infty} r^n (\cos n\theta + i \sin n\theta) &= \frac{1}{1 - r(\cos \theta + i \sin \theta)} = \frac{1}{(1 - r \cos \theta) - ir \sin \theta} \\ &= \frac{(1 - r \cos \theta) + ir \sin \theta}{(1 - r \cos \theta)^2 + (r \sin \theta)^2} = \frac{(1 - r \cos \theta) + ir \sin \theta}{1 - 2r \cos \theta + r^2}. \end{aligned}$$

上述等式的实部和虚部分别相等, 从而给出

$$\begin{aligned} \sum_{n=0}^{\infty} r^n \cos n\theta &= \frac{1 - r \cos \theta}{1 - 2r \cos \theta + r^2}, \quad 0 < r < 1, \\ \sum_{n=0}^{\infty} r^n \sin n\theta &= \frac{r \sin \theta}{1 - 2r \cos \theta + r^2}, \quad 0 < r < 1. \quad \square \end{aligned}$$

习题 1.16 下列函数有那些奇点, 如果是极点, 指出它的阶

$$\begin{aligned} (1) & \frac{1}{z^3 - z^2 - z + 1}, \quad (2) \frac{\ln(z+1)}{z}, \\ (3) & \frac{1}{\sin z^2}, \quad (4) \frac{1}{(1+z^2)(1+e^{\pi z})}. \end{aligned}$$

解：(1) 由于

$$\frac{1}{z^3 - z^2 - z + 1} = \frac{1}{(z-1)^2(z+1)},$$

因此 $f(z) = \frac{1}{z^3 - z^2 - z + 1}$ 有两个奇点, 分别是一阶极点 $z = -1$ 和二阶极点 $z = 1$.

(2) 函数 $f(z) = \frac{\ln(1+z)}{z}$ 在 $0 < |z| < 1$ 内的洛朗级数展开为:

$$\frac{\ln(1+z)}{z} = \frac{\sum_{k=1}^{\infty} (-1)^{k+1} \frac{z^k}{k}}{z} = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{z^{k-1}}{k} = \sum_{k=0}^{\infty} (-1)^k \frac{z^k}{k+1}.$$

因此 $z = 0$ 是函数 $f(z) = \frac{\ln(1+z)}{z}$ 的可去奇点。另一方面, $z = -1$ 是函数 $f(z) = \frac{\ln(1+z)}{z}$ 的支点 (也是奇点)。

(3) 函数 $f(z) = \frac{1}{\sin z^2}$ 的奇点为满足 $\sin z^2 = 0$ 的点。因此 $z_k = \sqrt{k\pi}$, $k \in \mathbb{Z}$ 均为函数 $f(z) = \frac{1}{\sin z^2}$ 的奇点 (当 $k \neq 0$ 时, $\sqrt{\cdot}$ 考虑多值情况)。进一步的, 当 $k = 0$ 时,

$$\lim_{z \rightarrow z_k} \frac{\sin z^2}{(z - z_k)^2} = \lim_{z \rightarrow z_k} \frac{z \cos z^2}{z - z_k} = \lim_{z \rightarrow z_k} (-2z^2 \sin z^2 + \cos z^2) = 1,$$

因此 $z_0 = 0$ 为二阶极点。而 $k \neq 0$ 时

$$\lim_{z \rightarrow z_k} \frac{\sin z^2}{z - z_k} = \lim_{z \rightarrow z_k} z \cos z^2 = (-1)^k z_k,$$

因此这些奇点均为单极点。

(4) 对于函数

$$f(z) = \frac{1}{(1+z^2)(1+e^{\pi z})} = \frac{1}{(1+zi)(1-zi)(1+e^{\pi z})},$$

易见 $z_k = i(2k+1)$, $k \in \mathbb{Z}$ 均为奇点。特别的, 当 $z = \pm i$ 均为二阶奇点, 而 $z = i(2k+1)$, $k \neq 0, 1$, $k \in \mathbb{Z}$ 为单极点。□

习题 1.17 确定下列函数的奇点, 求出函数在各孤立奇点处的留数

$$\begin{aligned} (1) \frac{e^z}{1+z}, \quad (2) \frac{e^{iz}}{z^2+a^2}, \quad (a>0), \quad (3) \frac{1}{z^3-z^5}, \quad (4) \frac{z^{2n}}{(z+1)^n}, \\ (5) e^{1/(1-z)}, \quad (6) \frac{1}{\sin \frac{1}{z}}, \quad (7) \frac{1}{z \sin z}, \quad (8) \frac{z^3}{(\cos z - 1)^2}. \end{aligned}$$

解: (1) 函数 $f(z) = \frac{e^z}{1+z}$ 的有单极点 $z_0 = -1$, 且

$$\text{Res } f(z_0) = \lim_{z \rightarrow -1} \frac{(1+z)e^z}{1+z} = e^{-1}.$$

(2) 我们有

$$f(z) = \frac{e^{iz}}{z^2+a^2} = \frac{e^{iz}}{(z+ai)(z-ai)},$$

因此函数 $f(z) = \frac{e^{iz}}{z^2+a^2}$ 具有两个单极点 $z_1 = ia$ 和 $z_2 = -ia$, 且有

$$\begin{aligned} \text{Res } f(z_1) &= \lim_{z \rightarrow ia} \frac{(z-ai)e^{iz}}{z^2+a^2} = \frac{e^{-a}}{ai+ai} = -\frac{ie^{-a}}{2a}, \\ \text{Res } f(z_2) &= \lim_{z \rightarrow -ia} \frac{(z+ai)e^{iz}}{z^2+a^2} = \frac{e^a}{-ai-ai} = \frac{ie^a}{2a}. \end{aligned}$$

(3) 我们有

$$f(z) = \frac{1}{z^3-z^5} = \frac{1}{z^3(1-z^2)} = \frac{1}{z^3(1-z)(1+z)},$$

因此函数 $f(z) = \frac{1}{z^3-z^5}$ 有单极点 $z_1 = 1$, $z_2 = -1$ 和三阶极点 $z_3 = 0$, 且有

$$\begin{aligned} \text{Res } f(z_1) &= \lim_{z \rightarrow 1} \frac{z-1}{z^3-z^5} = -\lim_{z \rightarrow 1} \frac{1}{z^3(1+z)} = -\frac{1}{2}, \\ \text{Res } f(z_2) &= \lim_{z \rightarrow -1} \frac{z+1}{z^3-z^5} = \lim_{z \rightarrow -1} \frac{1}{z^3(1-z)} = -\frac{1}{2}, \\ \text{Res } f(z_3) &= \frac{1}{2!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} \left(\frac{z^3}{z^3-z^5} \right) = \frac{1}{2!} \lim_{z \rightarrow 0} \frac{d^2}{dz^2} \left(\frac{1}{1-z^2} \right) = 1. \end{aligned}$$

(4) 函数 $f(z) = \frac{z^{2n}}{(z+1)^n}$ 具有 n 阶极点 $z_0 = -1$, 且有

$$\begin{aligned}\operatorname{Res} f(z_0) &= \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \frac{d^{n-1}}{dz^{n-1}} \left(\frac{(z+1)^n z^{2n}}{(z+1)^n} \right) = \frac{1}{(n-1)!} \lim_{z \rightarrow -1} \frac{d^{n-1}}{dz^{n-1}} (z^{2n}) \\ &= \frac{(2n)!}{(n-1)!(n+1)!} (-1)^{n+1}.\end{aligned}$$

(5) 函数 $f(z) = e^{\frac{1}{1-z}}$ 有本性奇点 $z_0 = 1$, 根据洛朗展开

$$e^{\frac{1}{1-z}} = \sum_{k=0}^{\infty} \frac{1}{k!} \left(\frac{1}{1-z} \right)^k = \sum_{k=-\infty}^0 \frac{(-1)^{-k} (z-1)^k}{(-k)!},$$

因此

$$\operatorname{Res} f(z_0) = -1. \quad (1.3)$$

(6) 函数 $f(z) = \frac{1}{\sin \frac{1}{z}}$ 有无数个单极点 $z_k = \frac{1}{k\pi}, k \in \mathbb{Z} \setminus \{0\}$, 以及非孤立奇点 $z_0 = 0$. 对于孤立奇点 z_k , 我们有

$$\operatorname{Res} f(z_k) = \lim_{z \rightarrow \frac{1}{k\pi}} \frac{z - \frac{1}{k\pi}}{\sin \frac{1}{z}} = \lim_{z \rightarrow \frac{1}{k\pi}} \frac{1}{-\frac{1}{z^2} \cos \frac{1}{z}} = \frac{(-1)^{k+1}}{k^2 \pi^2}.$$

(7) 函数 $f(z) = \frac{1}{z \sin z}$ 有单极点 $z_k = k\pi, k \in \mathbb{Z} \setminus \{0\}$ 和二阶极点 $z_0 = 0$, 且有

$$\begin{aligned}\operatorname{Res} f(z_0) &= \lim_{z \rightarrow 0} \frac{d}{dz} \left(\frac{z^2}{z \sin z} \right) = \lim_{z \rightarrow 0} \frac{\sin z - z \cos z}{\sin^2 z} = \lim_{z \rightarrow 0} \frac{z \sin z}{2 \sin z \cos z} = 0, \\ \operatorname{Res} f(z_k) &= \lim_{z \rightarrow k\pi} \frac{z - k\pi}{z \sin z} = \lim_{z \rightarrow k\pi} \frac{1}{\sin z + z \cos z} = \frac{(-1)^k}{k\pi}.\end{aligned}$$

(8) 我们有

$$f(z) = \frac{z^3}{(\cos z - 1)^2} = \frac{z^3}{(2 \sin^2 \frac{z}{2})^2} = \frac{z^3}{4 \sin^4 \frac{z}{2}},$$

因此函数 $f(z)$ 有单极点 $z_0 = 0$, 以及四阶极点 $z_k = 2k\pi, k \in \mathbb{Z} \setminus \{0\}$, 且有

$$\begin{aligned}\operatorname{Res} f(z_0) &= \lim_{z \rightarrow 0} \frac{z^4}{4 \sin^4 \frac{z}{2}} = 4, \\ \operatorname{Res} f(z_k) &= \frac{1}{3!} \lim_{z \rightarrow 2k\pi} \frac{d^3}{dz^3} \left(\frac{(z - 2k\pi)^4 z^3}{4 \sin^4 \frac{z}{2}} \right) = 4 + 8k^2 \pi^2. \quad \square\end{aligned}$$

习题 1.18 计算下列回路积分

$$\begin{aligned}(1) & \oint_{\ell} \frac{dz}{(z^2+1)(z-1)^2}, \quad (\ell \text{ 的方程是 } x^2 + y^2 - 2x - 2y = 0), \\ (2) & \oint_{|z|=2} \frac{z dz}{\frac{1}{2} - \sin^2 z}, \quad (3) \oint_{|z|=2} \frac{e^{2z}}{(z-1)^2} dz, \quad (4) \oint_{|z-i|=1} \frac{e^z - e^{-z}}{e^z + e^{-z}} dz.\end{aligned}$$

解: (1) 函数

$$f(z) = \frac{1}{(z^2+1)(z-1)^2} = \frac{1}{(z+i)(z-i)(z-1)^2},$$

有二阶极点 $z_1 = 1$, 单极点 $z_2 = i$ 和单极点 $z_3 = -i$ 。其中, z_1 和 z_2 在曲线 ℓ 所围区域的内部。进一步的

$$\begin{aligned}\operatorname{Res} f(z_1) &= \lim_{z \rightarrow 1} \frac{d}{dz} \left(\frac{(z-1)^2}{(z^2+1)(z-1)^2} \right) = \lim_{z \rightarrow 1} \frac{-2z}{(1+z^2)^2} = -\frac{1}{2}, \\ \operatorname{Res} f(z_2) &= \lim_{z \rightarrow i} \frac{(z-i)}{(z^2+1)(z-1)^2} = \frac{1}{4}.\end{aligned}$$

由此可得

$$\oint_{\ell} \frac{dz}{(z^2+1)(z-1)^2} = 2\pi i (\operatorname{Res} f(z_1) + \operatorname{Res} f(z_2)) = 2\pi i \left(\frac{1}{4} - \frac{1}{2} \right) = -\frac{1}{2}\pi i.$$

(2) 函数

$$f(z) = \frac{z}{\frac{1}{2} - \sin^2 z} = \frac{z}{(\frac{\sqrt{2}}{2} - \sin z)(\frac{\sqrt{2}}{2} + \sin z)},$$

具有一系列单极点 $z_k = \frac{(2k+1)\pi}{4}$, $k \in \mathbb{Z}$ 。在区域 $|z| \leq 2$ 内部, 有两个单极点 z_0 和 z_{-1} , 以及

$$\begin{aligned}\operatorname{Res} f(z_0) &= \lim_{z \rightarrow \frac{\pi}{4}} \frac{z(z - \frac{\pi}{4})}{\frac{1}{2} - \sin^2 z} = \lim_{z \rightarrow \frac{\pi}{4}} \frac{2z - \frac{\pi}{4}}{-2 \sin z \cos z} = -\frac{\pi}{4}, \\ \operatorname{Res} f(z_{-1}) &= \lim_{z \rightarrow -\frac{\pi}{4}} \frac{z(z + \frac{\pi}{4})}{\frac{1}{2} - \sin^2 z} = \lim_{z \rightarrow -\frac{\pi}{4}} \frac{2z + \frac{\pi}{4}}{-2 \sin z \cos z} = -\frac{\pi}{4}.\end{aligned}$$

由此可得

$$\oint_{|z|=2} \frac{zdz}{\frac{1}{2} - \sin^2 z} = 2\pi i (\operatorname{Res} f(z_0) + \operatorname{Res} f(z_{-1})) = -\pi^2 i.$$

(3) 函数 $f(z) = \frac{e^{2z}}{(z-1)^2}$ 有二阶奇点 $z_0 = 1$, 且其在区域 $|z| \leq 2$ 内部, 因此

$$\oint_{|z|=2} \frac{e^{2z}}{(z-1)^2} dz = 2\pi i \operatorname{Res} f(z_0) = 2\pi i \lim_{z \rightarrow 1} \frac{d}{dz} \left(\frac{(z-1)^2 e^{2z}}{(z-1)^2} \right) = 4\pi e^2 i.$$

(4) 函数 $f(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$ 有无穷个单极点 $z_k = \frac{2k+1}{2}\pi i$, $k \in \mathbb{Z}$ 。只有点 z_0 在区域 $|z - i| \leq 1$ 内部, 因此

$$\begin{aligned}\oint_{|z-i|=1} \frac{e^z - e^{-z}}{e^z + e^{-z}} dz &= 2\pi i \operatorname{Res} f(z_0) = 2\pi i \lim_{z \rightarrow \frac{1}{2}\pi i} \frac{(z - \frac{1}{2}\pi i)(e^z - e^{-z})}{e^z + e^{-z}} \\ &= 2\pi i \lim_{z \rightarrow \frac{1}{2}\pi i} \frac{(z - \frac{1}{2}\pi i)(e^z + e^{-z}) + (e^z - e^{-z})}{e^z - e^{-z}} = 2\pi i. \quad \square\end{aligned}$$

1.5 第 05 次课作业

习题 1.19 已知

$$\begin{aligned}I_1 &= \int_0^{\infty} f(x) \cos mx dx, \\ I_2 &= \int_0^{\infty} g(x) \sin mx dx.\end{aligned}$$

其中 $f(x)$ 和 $g(x)$ 为偶函数和奇函数, 它们在实轴上没有奇点, 在全平面除有限个奇点外是解析的; 当 z 在全平面 (含实轴) 上趋于无穷时, $f(z)$ 及 $g(z)$ 一致的 $\rightarrow 0$ 。对于 $m < 0$, 试用留数定理计算出 I_1 和 I_2 的积分主值计算公式 (注: 需给出完整的推导过程, 不能直接套用讲义中给出的结论)。

解: 由于 $f(x)$ 和 $g(x)$ 为偶函数和奇函数, 故有:

$$I_1 = \int_0^\infty f(x) \cos mx dx = \int_{-\infty}^\infty \frac{1}{2} f(x) e^{imx} dx,$$

$$I_2 = \int_0^\infty g(x) \sin mx dx = \int_{-\infty}^\infty \frac{1}{2i} g(x) e^{imx} dx.$$

考虑约当引理在 $m < 0$ 时的情况, 取 C_R 为以零点为圆心 R 为半径的半圆 (下半平面) 积分路径为逆时针方向, 我们有估计

$$\begin{aligned} \left| \int_{C_R} f(z) e^{imz} dz \right| &= \left| \int_{-\pi}^0 f(Re^{i\varphi}) e^{mR(i \cos \varphi - \sin \varphi)} Re^{i\varphi} i d\varphi \right| \\ &\leq \int_{-\pi}^0 |f(Re^{i\varphi})| e^{mR(i \cos \varphi - \sin \varphi)} R e^{i\varphi} |i| d\varphi = \int_{-\pi}^0 |f(Re^{i\varphi})| e^{-mR \sin \varphi} R d\varphi \\ &\leq \max |f(z)| \int_{-\pi}^0 e^{-mR \sin \varphi} R d\varphi = \max |f(z)| \int_0^\pi e^{mR \sin \varphi} R d\varphi \\ &= 2 \max |f(z)| \int_0^{\pi/2} e^{mR \sin \varphi} R d\varphi \rightarrow 0 \quad (R \rightarrow \infty). \end{aligned}$$

因此

$$\int_{-R}^{-R} \frac{1}{2} f(x) e^{imx} dx + \int_{C_R} \frac{1}{2} f(z) e^{imz} dz = \pi i \sum_{z_0 \in \{\text{半圆弧 } C_R \text{ 和线段 } [-R, R] \text{ 包围区域内的孤立奇点全体}\}} \text{Res} (f(z_0) e^{imz_0}).$$

取 $R \rightarrow \infty$, 则有

$$I_1 = \int_{-\infty}^\infty \frac{1}{2} f(x) e^{imx} dx = -\pi i \sum_{z_0 \in \{\text{下半平面内奇点全体}\}} \text{Res} (f(z_0) e^{imz_0}).$$

类似的,

$$I_2 = \int_{-\infty}^\infty \frac{1}{2i} g(x) e^{imx} dx = -\pi \sum_{z_0 \in \{\text{下半平面内奇点全体}\}} \text{Res} (g(z_0) e^{imz_0}). \quad \square$$

习题 1.20 计算下列实变函数定积分

$$\begin{aligned} (1) \int_0^{2\pi} \frac{dx}{(1 + \varepsilon \cos x)^2}, \quad 0 < \varepsilon < 1, \quad (2) \int_0^{2\pi} \frac{\sin^2 x dx}{a + b \cos x}, \quad (a > b > 0), \\ (3) \int_0^{2\pi} \frac{\cos x dx}{1 - 2\varepsilon \cos x + \varepsilon^2}, \quad (|\varepsilon| < 1), \quad (4) \int_0^{2\pi} \cos^{2n} x dx, \\ (5) \int_0^{2\pi} \frac{dx}{2 + \cos x}, \quad (6) \int_0^\pi \frac{a dx}{a^2 + \sin^2 x}, \quad (a > 0). \end{aligned}$$

解: (1) 设 $z = e^{ix}$, 则有:

$$\begin{aligned} I &= \int_0^{2\pi} \frac{dx}{(1 + \varepsilon \cos x)^2} = \oint_{|z|=1} \frac{\frac{dz}{iz}}{(1 + \frac{\varepsilon}{2}(z + \frac{1}{z}))^2} = \oint_{|z|=1} \frac{-4zidz}{(\varepsilon z^2 + 2z + \varepsilon)^2} \\ &= \oint_{|z|=1} \frac{-4zidz}{\varepsilon^2(z - \frac{-1+\sqrt{1-\varepsilon^2}}{\varepsilon})^2(z + \frac{1+\sqrt{1-\varepsilon^2}}{\varepsilon})^2} = \frac{8\pi}{\varepsilon^2} \text{Res } f(z_0), \end{aligned}$$

上式中 (只有 z_0 在 $|z|=1$ 所围区域内部)

$$f(z) = \frac{z}{(z - \frac{-1+\sqrt{1-\varepsilon^2}}{\varepsilon})^2(z + \frac{1+\sqrt{1-\varepsilon^2}}{\varepsilon})^2}, \quad z_0 = \frac{-1 + \sqrt{1 - \varepsilon^2}}{\varepsilon}.$$

因此

$$I = \frac{8\pi}{\varepsilon^2} \lim_{z \rightarrow z_0} \frac{d}{dz} ((z - z_0)^2 f(z)) = \frac{8\pi}{\varepsilon^2} \cdot \frac{\varepsilon^2}{4(1 - \varepsilon^2)^{3/2}} = \frac{2\pi}{(1 - \varepsilon^2)^{3/2}}.$$

(2) 设 $z = e^{ix}$, 则有

$$I = \int_0^{2\pi} \frac{\sin^2 x dx}{a + b \cos x} = \oint_{|z|=1} \frac{-\frac{1}{4}(z - \frac{1}{z})^2 \frac{dz}{iz}}{a + \frac{b}{2}(z + \frac{1}{z})} = \oint_{|z|=1} \frac{i}{2} \frac{(z-1)^2(z+1)^2}{(bz^2 + 2az + b)z^2} = \frac{i}{2} \oint_{|z|=1} f(z) dz$$

这里函数

$$f(z) = \frac{(z-1)^2(z+1)^2}{(bz^2 + 2az + b)z^2},$$

它有二阶极点 $z_0 = 0$ 和两个单极点

$$z_1 = -\frac{a}{b} + \sqrt{\frac{a^2}{b^2} - 1}, \quad z_2 = -\frac{a}{b} - \sqrt{\frac{a^2}{b^2} - 1}.$$

而只有点 z_0 和 z_1 在 $|z|=1$ 所围区域内部, 因此

$$\begin{aligned} I &= -\pi (\text{Res } f(z_0) + \text{Res } f(z_1)) = -\pi \lim_{z \rightarrow z_0} \frac{d}{dz} ((z - z_0)^2 f(z)) - \pi \lim_{z \rightarrow z_1} ((z - z_1) f(z)) \\ &= -\pi \left(-\frac{2a}{b^2} + \frac{2}{b} \sqrt{\frac{a^2}{b^2} - 1} \right) = \frac{2\pi(a - \sqrt{a^2 - b^2})}{b^2}. \end{aligned}$$

(3) 设 $z = e^{ix}$, 则有

$$I = \int_0^{2\pi} \frac{\cos x dx}{1 - 2\varepsilon \cos x + \varepsilon^2} = \oint_{|z|=1} \frac{\frac{1}{2}(z + \frac{1}{z}) \frac{dz}{iz}}{1 - \varepsilon(z + \frac{1}{z}) + \varepsilon^2} = \oint_{|z|=1} -\frac{i}{2} \frac{(z^2 + 1)dz}{z^2 - \varepsilon(z^3 + z) + \varepsilon^2 z^2} = \frac{i}{2} \oint_{|z|=1} f(z) dz.$$

这里函数

$$f(z) = \frac{z^2 + 1}{z(z - \varepsilon)(\varepsilon z - 1)},$$

它有三个单极点

$$z_0 = 0, \quad z_1 = \varepsilon, \quad z_2 = \frac{1}{\varepsilon},$$

而只有点 z_0 和 z_1 在 $|z| = 1$ 所围区域内部, 因此

$$\begin{aligned} I &= -\pi (\operatorname{Res} f(z_0) + \operatorname{Res} f(z_1)) = -\pi \left(\lim_{z \rightarrow z_0} ((z - z_0)f(z)) + \lim_{z \rightarrow z_1} ((z - z_1)f(z)) \right) \\ &= -\pi \left(\frac{1}{\varepsilon} + \frac{\varepsilon^2 + 1}{\varepsilon(\varepsilon^2 - 1)} \right) = \frac{2\pi\varepsilon}{1 - \varepsilon^2}. \end{aligned}$$

(4) 设 $z = e^{ix}$, 则有

$$I = \int_0^{2\pi} \cos^{2n} x dx = \oint_{|z|=1} \frac{1}{2^{2n}} \left(z + \frac{1}{z} \right)^{2n} \frac{dz}{iz} = -\frac{i}{2^{2n}} \oint_{|z|=1} f(z) dz.$$

这里函数

$$f(z) = \frac{(z^2 + 1)^{2n}}{z^{2n+1}},$$

它有一个单极点 $z_0 = 0$, 且在 $|z| = 1$ 所围区域内部, 因此

$$\begin{aligned} I &= \frac{\pi}{2^{2n-1}} \operatorname{Res} f(z_0) = \frac{\pi}{2^{2n-1}} \frac{1}{(2n)!} \lim_{z \rightarrow z_0} \frac{d^{2n}}{dz^{2n}} ((z - z_0)^{2n+1} f(z)) \\ &= \frac{\pi}{2^{2n-1}} \frac{1}{(2n)!} \lim_{z \rightarrow z_0} \frac{d^{2n}}{dz^{2n}} ((z^2 + 1)^{2n}) = \frac{\pi}{2^{2n-1}} \frac{1}{(2n)!} C_{2n}^n (2n)! = \frac{\pi}{2^{2n-1}} C_{2n}^n = 2\pi \frac{(2n-1)!!}{2n!!}. \end{aligned}$$

其中, 上式中

$$\lim_{z \rightarrow z_0} \frac{d^{2n}}{dz^{2n}} ((z^2 + 1)^{2n}),$$

的计算, 只需要考虑多项式 $((z^2 + 1)^{2n})$ 中 z^{2n} 的系数即可, 它是 C_{2n}^n . $\square$

(5) 设 $z = e^{ix}$, 则有

$$I = \int_0^{2\pi} \frac{dx}{2 + \cos x} = \oint_{|z|=1} \frac{\frac{dz}{iz}}{2 + \frac{1}{2} \left(z + \frac{1}{z} \right)} = -2i \oint_{|z|=1} f(z) dz.$$

这里函数

$$f(z) = \frac{1}{z^2 + 4z + 1},$$

它有两个单极点

$$z_1 = -2 + \sqrt{3}, \quad z_2 = -2 - \sqrt{3},$$

只有点 z_1 在 $|z| = 1$ 所围区域内部, 因此

$$I = 4\pi \operatorname{Res} f(z_1) = 4\pi \lim_{z \rightarrow z_1} ((z - z_1)f(z)) = \frac{4\pi}{2\sqrt{3}} = \frac{2\sqrt{3}\pi}{3}.$$

(6) 设 $z = e^{ix}$, 则有

$$\begin{aligned} I &= \int_0^\pi \frac{a dx}{a^2 + \sin^2 x} = \int_0^\pi \frac{2a dx}{2a^2 + 1 - \cos 2x} = \int_0^{2\pi} \frac{a dx}{2a^2 + 1 - \cos x} \\ &= \oint_{|z|=1} \frac{a \frac{dz}{iz}}{2a^2 + 1 - \frac{1}{2} \left(z + \frac{1}{z} \right)} = 2ai \oint_{|z|=1} f(z) dz. \end{aligned}$$

这里函数

$$f(z) = \frac{1}{z^2 - 2(2a^2 + 1)z + 1},$$

它有两个单极点

$$z_1 = (2a^2 + 1) - 2a\sqrt{a^2 + 1}, \quad z_2 = (2a^2 + 1) + 2a\sqrt{a^2 + 1},$$

只有点 z_1 在 $|z| = 1$ 所围区域内部, 因此

$$I = -4\pi a \operatorname{Res} f(z_1) = -4\pi a \lim_{z \rightarrow z_1} ((z - z_1)f(z)) = \frac{\pi}{\sqrt{a^2 + 1}}. \quad \square$$

习题 1.21 计算下列实变函数定积分

$$\begin{aligned} (1) & \int_{-\infty}^{\infty} \frac{dx}{(x^2 + a^2)^2(x^2 + b^2)}, \quad (a, b > 0), \quad (2) \int_{-\infty}^{\infty} \frac{1}{x(x-1)(x-3)} dx, \\ (3) & \int_0^{\infty} \frac{x^2 + 1}{x^6 + 1} dx, \quad (4) \int_0^{\infty} \frac{x^{2m}}{x^{2n} + 1} dx, \quad (m < n), \quad (5) \int_0^{\infty} \frac{x^2}{(x^2 + a^2)^2} dx, \quad (a > 0). \end{aligned}$$

解: (1) 该积分属于类型二, 考虑函数

$$f(z) = \frac{1}{(z^2 + a^2)^2(z^2 + b^2)} = \frac{1}{(z + ai)^2(z - ai)^2(z + bi)(z - bi)},$$

其在上半平面上有一个单极点 $z_1 = bi$ 和二阶极点 $z_2 = ai$, 以及

$$\begin{aligned} \operatorname{Res} f(z_1) &= \lim_{z \rightarrow z_1} ((z - z_1)f(z)) = -\frac{i}{2b(a^2 - b^2)^2}, \\ \operatorname{Res} f(z_2) &= \lim_{z \rightarrow z_2} \frac{d}{dz} ((z - z_2)^2 f(z)) = \lim_{z \rightarrow z_2} \frac{d}{dz} \left(\frac{1}{(z + ai)^2(z^2 + b^2)} \right) = \frac{(3a^2 - b^2)i}{4a^3(a^2 - b^2)^2}. \end{aligned}$$

因此有

$$I = 2\pi i (\operatorname{Res} f(z_1) + \operatorname{Res} f(z_2)) = \frac{\pi(2a^3 + b^3 - 3a^2b)}{2a^3b(b^2 - a^2)^2}.$$

(2) 该积分属于类型二、四, 考虑函数

$$f(z) = \frac{1}{z(z-1)(z-3)},$$

其在实轴上有单极点 $z_0 = 0$, $z_1 = 1$, $z_2 = 3$, 以及

$$\begin{aligned} \operatorname{Res} f(z_0) &= \lim_{z \rightarrow z_0} ((z - z_0)f(z)) = \frac{1}{3}, \\ \operatorname{Res} f(z_1) &= \lim_{z \rightarrow z_1} ((z - z_1)f(z)) = -\frac{1}{2}, \\ \operatorname{Res} f(z_2) &= \lim_{z \rightarrow z_2} ((z - z_2)f(z)) = \frac{1}{6}. \end{aligned}$$

因此有

$$I = \pi i \sum_{k=0}^2 \operatorname{Res} f(z_k) = 0.$$

(3) 首先有

$$I = \int_0^\infty \frac{x^2 + 1}{x^6 + 1} dx = \frac{1}{2} \int_{-\infty}^\infty \frac{1}{x^4 - x^2 + 1} dx,$$

该积分属于类型二。考虑函数

$$f(z) = \frac{1}{z^4 - z^2 + 1},$$

有四个单极点

$$z_0 = e^{\frac{1}{6}\pi i}, z_2 = e^{\frac{5}{6}\pi i}, z_3 = e^{\frac{7}{6}\pi i}, z_5 = e^{\frac{11}{6}\pi i}.$$

其中 z_0 和 z_2 在上半平面, 计算 $f(z)$ 在这些点的留数, 可得

$$\begin{aligned} \operatorname{Res} f(z_0) &= \lim_{z \rightarrow z_0} ((z - z_0)f(z)) = \frac{1}{(z_0 - z_2)(z_0 - z_3)(z_0 - z_5)} = -\frac{\sqrt{3} + 3i}{12}, \\ \operatorname{Res} f(z_2) &= \lim_{z \rightarrow z_2} ((z - z_2)f(z)) = \frac{1}{(z_2 - z_0)(z_2 - z_3)(z_2 - z_5)} = \frac{\sqrt{3} - 3i}{12}. \end{aligned}$$

因此有

$$I = \pi i (\operatorname{Res} f(z_0) + \operatorname{Res} f(z_2)) = \frac{1}{2} \pi.$$

(4) 首先有

$$I = \int_0^\infty \frac{x^{2m}}{x^{2n} + 1} dx = \frac{1}{2} \int_{-\infty}^\infty \frac{x^{2m}}{x^{2n} + 1} dx,$$

该积分属于类型二。考虑函数

$$f(z) = \frac{z^{2m}}{z^{2n} + 1},$$

有 $2n$ 个单极点

$$z_k = e^{\frac{(2k+1)\pi i}{2n}}, k = 0, 1, \dots, 2n-1.$$

其中 $z_0, z_1, \dots, z_{n-1}$ 在上半平面, 计算 $f(z)$ 在这些点的留数, 可得

$$\begin{aligned} \operatorname{Res} f(z_k) &= \lim_{z \rightarrow z_k} ((z - z_k)f(z)) = \lim_{z \rightarrow z_k} \frac{z^{2m} + 2mz^{2m-1}(z - z_k)}{2nz^{2n-1}} \\ &= \frac{1}{2n} z_k^{2(m-n)+1} = \frac{1}{2n} e^{\frac{(2(m-n)+1)(2k+1)\pi i}{2n}} = \frac{1}{2n} \eta^{k+\frac{1}{2}}, k = 0, 1, \dots, n-1, \end{aligned}$$

这里

$$\eta = e^{\frac{2(m-n)+1}{n}\pi i}.$$

因此有

$$\begin{aligned} I &= \pi i \sum_{k=0}^{n-1} \operatorname{Res} f(z_k) = \pi i \frac{\eta^{1/2} \eta^n - 1}{2n \eta - 1} = -\frac{\pi i}{n} \frac{e^{\frac{2(m-n)+1}{2n}\pi i}}{e^{\frac{2(m-n)+1}{n}\pi i} - 1} \\ &= -\frac{\pi i}{n} \frac{1}{e^{\frac{2(m-n)+1}{2n}\pi i} - e^{-\frac{2(m-n)+1}{2n}\pi i}} = \frac{\pi i}{n} \frac{1}{e^{\frac{2m+1}{2n}\pi i} - e^{-\frac{2m+1}{2n}\pi i}} = \frac{\pi}{2n \sin \frac{2m+1}{2n}\pi}. \end{aligned}$$

(5) 首先有

$$I = \int_0^\infty \frac{x^2}{(x^2 + a^2)^2} dx = \frac{1}{2} \int_{-\infty}^\infty \frac{x^2}{(x^2 + a^2)^2} dx$$

该积分属于类型二。考虑函数

$$f(z) = \frac{z^2}{(z^2 + a^2)^2},$$

在上半平面有一个二阶节点 $z_0 = ai$, 以及

$$\operatorname{Res} f(z_0) = \lim_{z \rightarrow z_0} \frac{d}{dz} ((z - z_0)^2 f(z)) = \lim_{z \rightarrow z_0} \frac{d}{dz} \left(\frac{z^2}{(z + ai)^2} \right) = -\frac{i}{4a}.$$

因此

$$I = \pi i \operatorname{Res} f(z_0) = \frac{\pi}{4a}. \square$$

习题 1.22 计算下列实变函数定积分

$$\begin{aligned} (1) & \int_0^\infty \frac{\cos x}{(x^2 + a^2)(x^2 + b^2)} dx, \quad (a, b > 0) & (2) & \int_{-\infty}^\infty \frac{e^{imx}}{x - i\alpha} dx, \quad (m > 0, \operatorname{Re} \alpha > 0), \\ (3) & \int_0^\infty \frac{\sin mx}{x(x^2 + a^2)} dx, \quad (m > 0, a > 0), & (4) & \int_0^\infty \frac{x \sin x}{1 + x^2} dx, \\ (5) & \int_0^\infty \frac{\cos mx}{1 + x^4} dx, \quad (m > 0), & (6) & \int_0^\infty \frac{\sin^2 x}{x^2} dx. \end{aligned}$$

解: (1) 首先有

$$I = \int_0^\infty \frac{\cos x}{(x^2 + a^2)(x^2 + b^2)} dx = \frac{1}{2} \int_{-\infty}^\infty \frac{\cos x}{(x^2 + a^2)(x^2 + b^2)} dx = \frac{1}{2} \int_{-\infty}^\infty \frac{e^{ix}}{(x^2 + a^2)(x^2 + b^2)} dx.$$

该积分属于类型三, 考虑函数

$$f(z) = \frac{e^{iz}}{(z^2 + a^2)(z^2 + b^2)},$$

在上半平面有两个单极点 $z_1 = ai$ 和 $z_2 = bi$, 以及

$$\operatorname{Res} f(z_1) = \lim_{z \rightarrow z_1} ((z - z_1)f(z)) = \frac{e^{-a}}{2ai(b^2 - a^2)},$$

$$\operatorname{Res} f(z_2) = \lim_{z \rightarrow z_2} ((z - z_2)f(z)) = \frac{e^{-b}}{2bi(a^2 - b^2)}.$$

因此

$$I = \pi i (\operatorname{Res} f(z_1) + \operatorname{Res} f(z_2)) = \frac{\pi(be^{-a} - ae^{-b})}{2ab(b^2 - a^2)}.$$

(2) 该积分属于类型三, 考虑函数

$$f(z) = \frac{e^{imz}}{z - i\alpha},$$

在上半平面有单极点 $z_0 = \alpha i$, 以及

$$\operatorname{Res} f(z_0) = \lim_{z \rightarrow z_1} ((z - z_1)f(z)) = e^{-m\alpha}.$$

因此

$$I = 2\pi i \operatorname{Res} f(z_0) = 2\pi i e^{-m\alpha}.$$

(3) 首先有

$$I = \int_0^\infty \frac{\sin mx}{x(x^2 + a^2)} dx = \frac{1}{2} \int_{-\infty}^\infty \frac{\sin mx}{x(x^2 + a^2)} dx = \frac{1}{2i} \int_{-\infty}^\infty \frac{e^{imx}}{x(x^2 + a^2)} dx$$

该积分属于类型三、四。考虑函数

$$f(z) = \frac{e^{imz}}{z(z^2 + a^2)},$$

其在实轴上有单极点 $z_0 = 0$, 在上半平面上有单极点 $z_1 = ai$, 以及

$$\begin{aligned} \operatorname{Res} f(z_0) &= \lim_{z \rightarrow z_0} ((z - z_0)f(z)) = \frac{1}{a^2}, \\ \operatorname{Res} f(z_1) &= \lim_{z \rightarrow z_1} ((z - z_1)f(z)) = -\frac{e^{-ma}}{2a^2}. \end{aligned}$$

因此

$$I = \frac{\pi}{2} \operatorname{Res} f(z_0) + \pi \operatorname{Res} f(z_1) = \frac{\pi}{2a^2} - \frac{\pi e^{-ma}}{2a^2}.$$

(4) 首先有

$$I = \int_0^\infty \frac{x \sin x}{1 + x^2} dx = \frac{1}{2} \int_{-\infty}^\infty \frac{x \sin x}{(1 + x^2)} dx = \frac{1}{2i} \int_{-\infty}^\infty \frac{x e^{ix}}{1 + x^2} dx,$$

该积分属于类型三。考虑函数

$$f(z) = \frac{z e^{iz}}{1 + z^2}$$

其在上半平面上有单极点 $z_1 = i$, 以及

$$\operatorname{Res} f(z_1) = \lim_{z \rightarrow z_1} ((z - z_1)f(z)) = \frac{1}{2} e^{-1}.$$

因此

$$I = \pi \operatorname{Res} f(z_1) = \frac{\pi}{2e}.$$

(5) 首先有

$$I = \int_0^\infty \frac{\cos mx}{1 + x^4} dx = \frac{1}{2} \int_{-\infty}^\infty \frac{\cos mx}{1 + x^4} dx = \frac{1}{2} \int_{-\infty}^\infty \frac{e^{imx}}{1 + x^4} dx,$$

该积分属于类型三。考虑函数

$$f(z) = \frac{e^{imz}}{1 + z^4},$$

其在上半平面有两个单极点 $z_1 = e^{i\pi/4}$ 和 $z_2 = e^{3i\pi/4}$, 以及

$$\begin{aligned} \operatorname{Res} f(z_1) &= \lim_{z \rightarrow z_1} ((z - z_1)f(z)) = \frac{e^{imz_1}}{4z_1^3} = \frac{1}{4} e^{-\sqrt{2}m/2} e^{i(\sqrt{2}m/2 - 3\pi/4)}, \\ \operatorname{Res} f(z_2) &= \lim_{z \rightarrow z_2} ((z - z_2)f(z)) = \frac{e^{imz_2}}{4z_2^3} = \frac{1}{4} e^{-\sqrt{2}m/2} e^{i(-\sqrt{2}m/2 - \pi/4)}. \end{aligned}$$

因此

$$\begin{aligned} I &= \pi i (\operatorname{Res} f(z_1) + \operatorname{Res} f(z_2)) = \frac{\pi i}{4} e^{-\sqrt{2}m/2} \left(e^{i(\sqrt{2}m/2 - 3\pi/4)} + e^{i(-\sqrt{2}m/2 - \pi/4)} \right) \\ &= -\frac{\pi}{2} e^{-\sqrt{2}m/2} \sin \left(\sqrt{2}m/2 - 3\pi/4 \right). \end{aligned}$$

(6) 首先有

$$I = \int_0^\infty \frac{\sin^2 x}{x^2} dx = \frac{1}{4} \int_{-\infty}^\infty \frac{1 - \cos 2x}{x^2} dx = \frac{1}{4} \int_{-\infty}^\infty \frac{1 - e^{2ix}}{x^2} dx.$$

该积分属于类型三、四。考虑函数

$$f(z) = \frac{1 - e^{2iz}}{z^2},$$

其在实轴上有单极点 $z_0 = 0$, 以及

$$\operatorname{Res} f(z_0) = \lim_{z \rightarrow z_0} ((z - z_0)f(z)) = -2ie^{iz}|_{z=0} = -2i.$$

因此

$$I = \frac{\pi i}{4} \operatorname{Res} f(z_0) = \frac{\pi}{2}. \quad \square$$